

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

yep no

fine all fine good

everything is good

Today's agenda

- Practical Task Discussion
- Total Recall
- RESTful
- Java Sockets
- Code examples
- Quiz
- Questions

Discuss obligatory practical task

Total Recall



What do you remember from the previous workshop ?

hibernate

get,post

5 versions of http

HTTP, methods, body,
etc

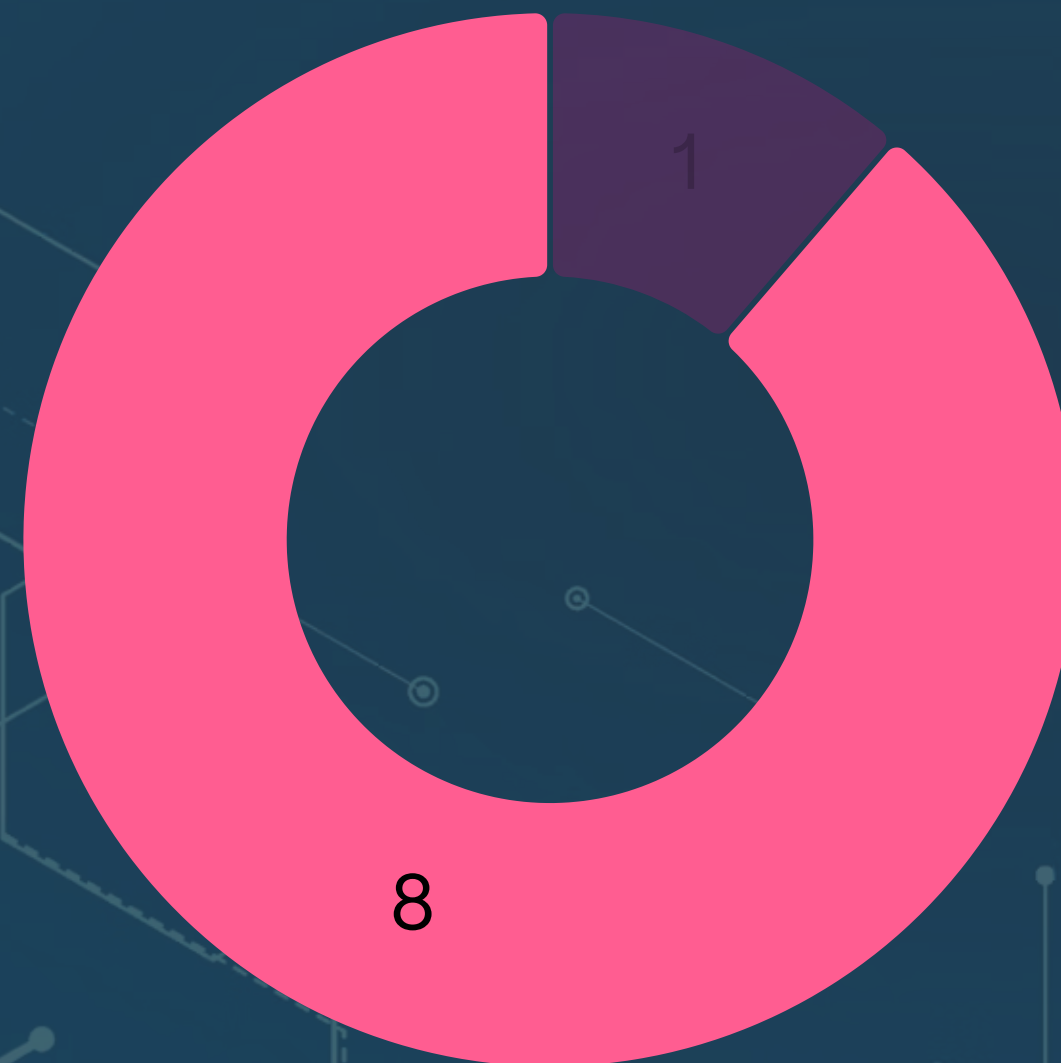
http versions

get post patch head
reset

http

about http

How many versions in HTTP ?



0	2	×
0	3	×
1	4	×
8	5	✓

ITPU IT PARK UNIVERSITY Http Versions

HTTP/0.9 - The first version, introduced in 1991.
Handling only basic GET requests.

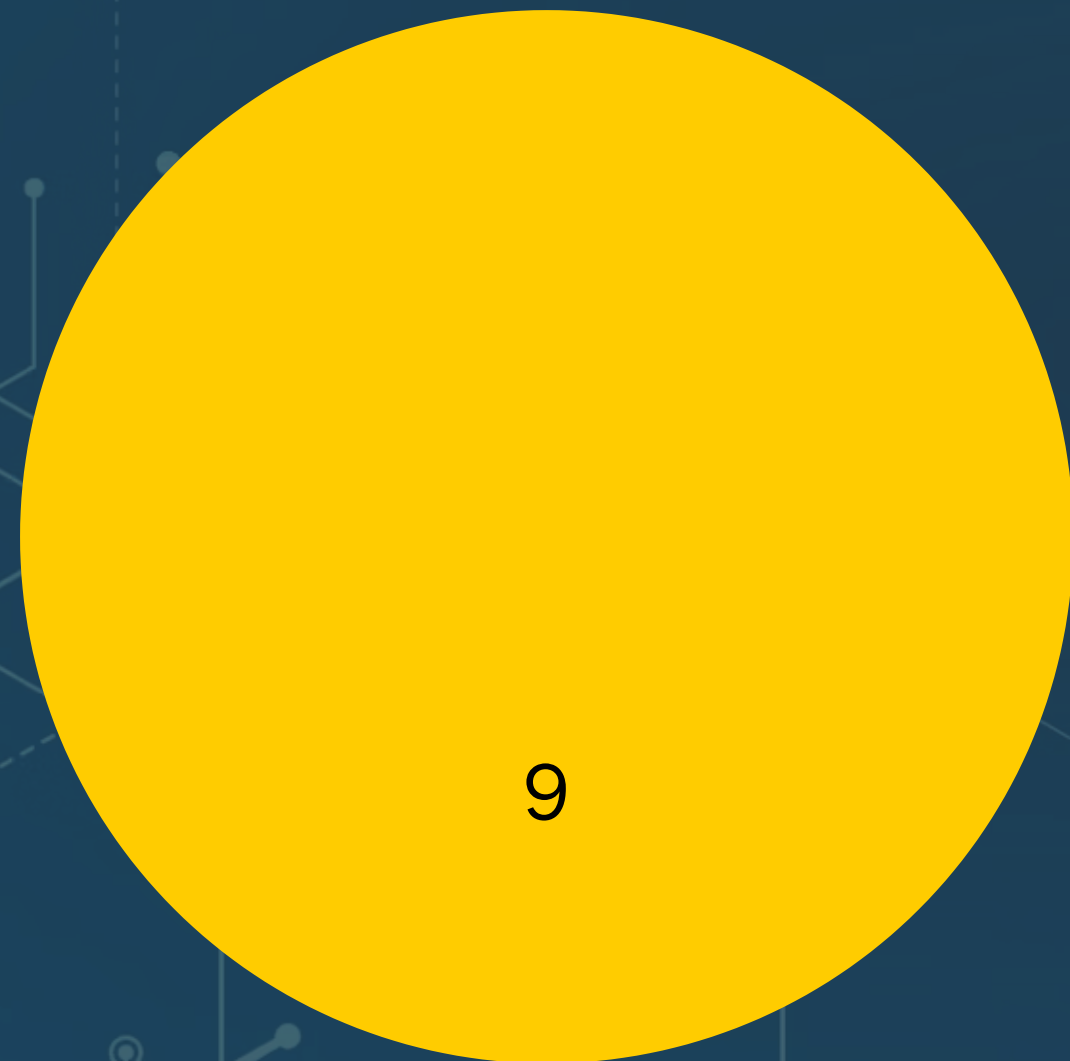
HTTP/1.0: Considered obsolete, rarely used
today.

HTTP/1.1: The long-standing standard version, still
widely implemented.

HTTP/2: Introduced significant performance
improvements over HTTP/1.1.

HTTP/3: The newest version, utilizing QUIC
protocol for faster connection establishment.

Which part of an HTTP request indicates the desired action to be performed on a specified resource?



- | | | | |
|----------------------------------|---|-------------------|----------------------------------|
| ☐ | 0 | A) Request URI | ☐ |
| ☒ | 9 | B) Request method | ☒ |
| ☐ | 0 | C) Request header | ☐ |
| ☐ | 0 | D) Request body | ☐ |

What is Idempotence ? What methods are idempotent ?

doing an operation once or doing it many times leaves the system in the same state.

ai is thinking

GET, HEAD, OPTIONS,
TRACE

get put delete

dunno

doing the exact same operation multiple times produces the same outcome as doing it just once

no

property of certain operations

What is Idempotence ? What methods are idempotent ?

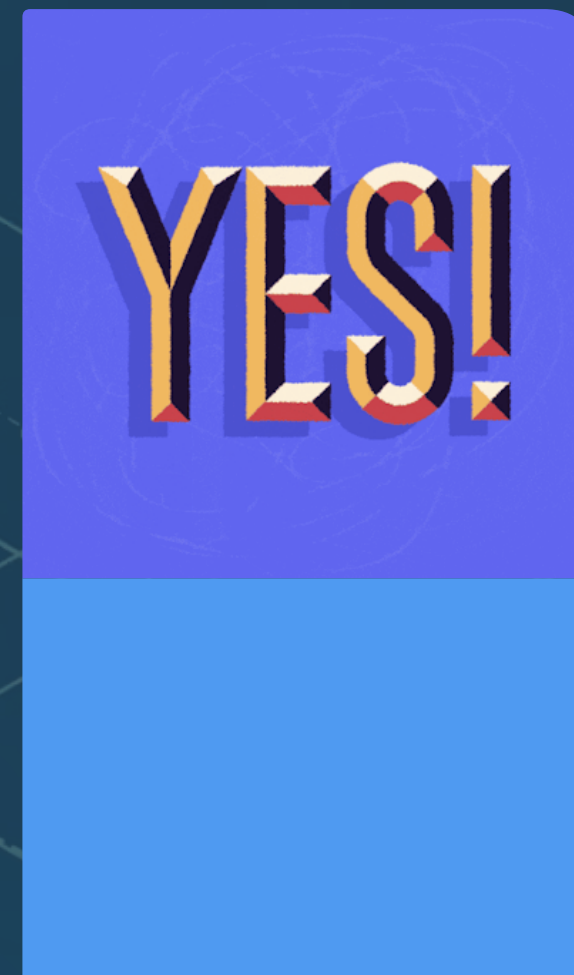
Idempotence is a property of an operation where performing it once or multiple times produces the same result.

get, delete, put -
duplicating the request
but getting same answer

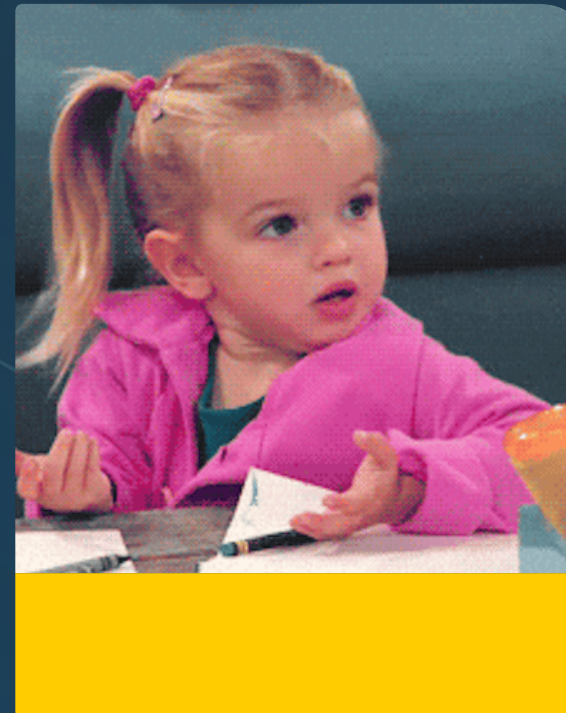
Http (ws-10)

RESTful | SOCKET





1



2



nooooooooo

REST

<https://restfulapi.net/richardson-maturity-model/>

<https://martinfowler.com/articles/richardsonMaturityModel.html>

Java Sockets

An endpoint for communication between
two machines

<https://www.baeldung.com/a-guide-to-java-sockets>

Quiz

Which HTTP method is used to fully update the current state of a resource?

17 ✓

1 ✗

A) POST

1 ✗

B) GET

C) PUT

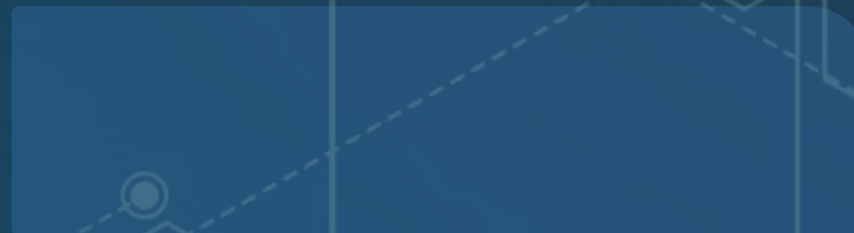
1 ✗

D) DELETE

What are the three main components of an HTTP response?

13 ✓

3 ✗

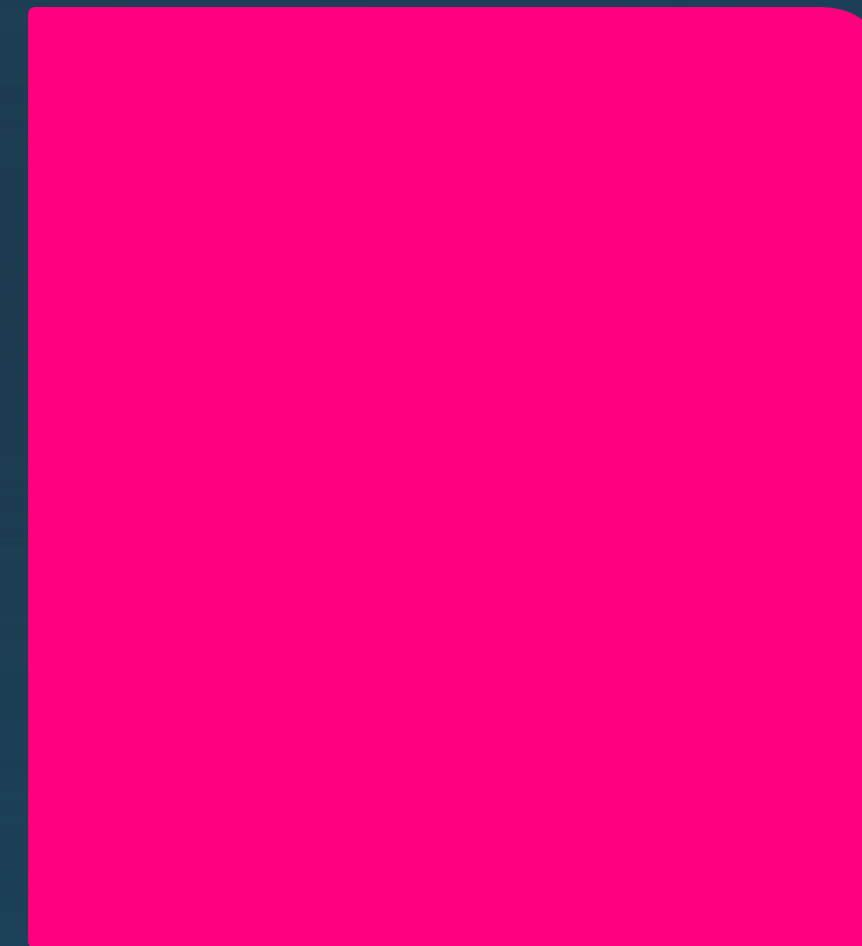


A) Status code, body, status line

3 ✗



B) body, method, status code



C) Status line, headers, body

1 ✗



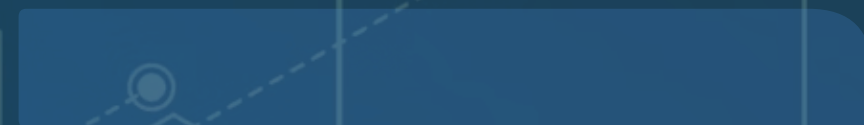
D) headers, status code, line

Which HTTP method is idempotent, meaning it can be called multiple times without different outcomes?

18 ✓



2 ✗



A) POST

B) GET

0 ✗

C) PATCH

0 ✗

D) DELETE

Which HTTP status code indicates that the requested resource is no longer available and no forwarding address is known?

22 ✓



0 ✗

A) 404 Not Found

0 ✗

C) 301 Moved Permanently

1 ✗

D) 503 Service Unavailable

B) 410 Gone

What does REST stand for?

21 ✓



0 ✗

A) Representational State Transfer

B) Real-time Elastic Systems
Transfer

2 ✗

C) Reliable Endpoint System
Technology

0 ✗

D) Regional Server Transfer

How many mature levels are there in Richardson's model ?

22 ✓



0 ✗

0 ✗

1 ✗

6

5

4

3

What is a common format for sending data in a RESTful API?

23 ✓

0 ✗

A) .exe

0 ✗

B) .png

0 ✗

D) .docx

C) JSON

What does it mean for a RESTful API to be stateless?

21 ✓



2 ✗

B) The server remembers each client between requests

1 ✗

C) The client must manage the state of the server

0 ✗

D) Resources must not be cached

A) The server doesn't store any state about the client session

Which Java class provides methods for reading and writing data through sockets?

4 ✗



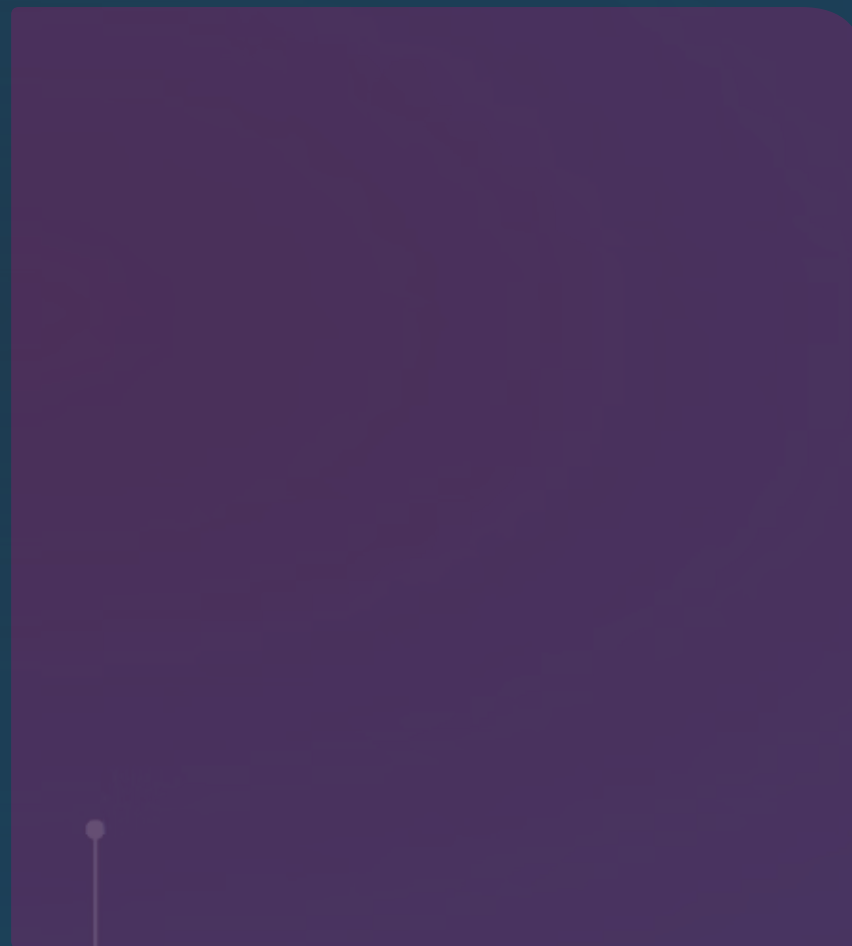
A) OutputStream

6 ✗



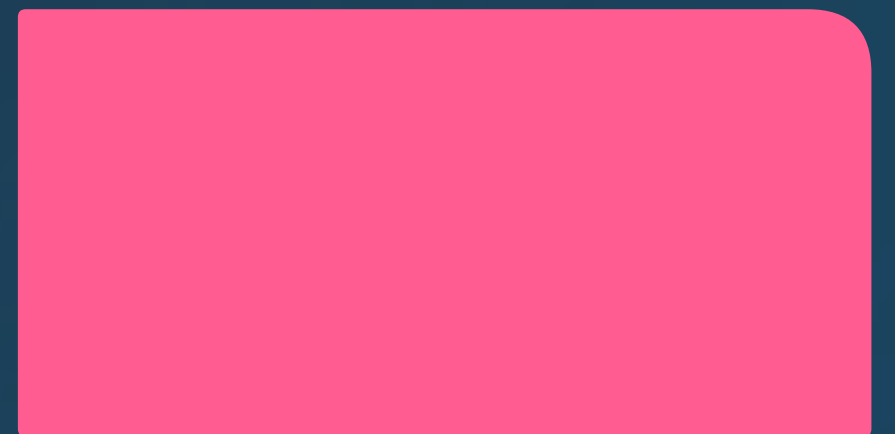
B) InputStream

9 ✗



C) SocketStream

4 ✓



D) DataStreams

Which class in Java is used for creating server sockets?

22 ✓

0 ✗

A) InetAddress

0 ✗

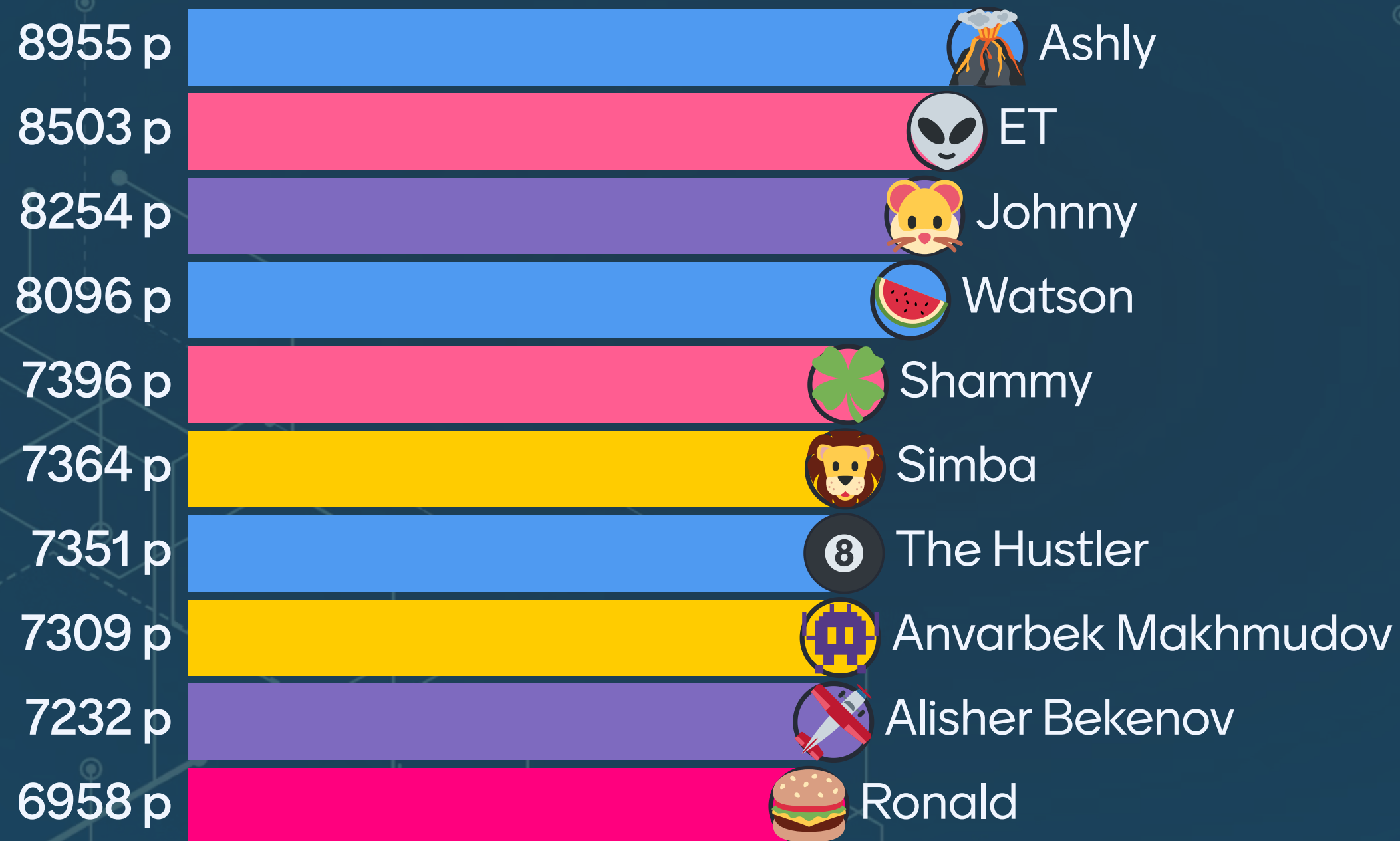
B) URLConnection

C) ServerSocket

2 ✗

D) SocketAddress

Quiz leaderboard



Next time: Servlet

Questions?

